

Bead-Pull Physics Chain

The reduced reconstruction starts from the measured complex reflection map, isolates the bead response in time, removes the slowly varying baseline, and converts the residual into a field-sensitive quantity.

Measured S_{11}

$S_{11}(x, y, z, \nu)$
Measured on the full aperture for each retained z plane.

Time Gate

$S_{11}^{\text{gated}}(t) = G(t - t_{\text{bead}})S_{11}(t)$
Keeps the bead return and rejects distant reflections.

Remove Baseline

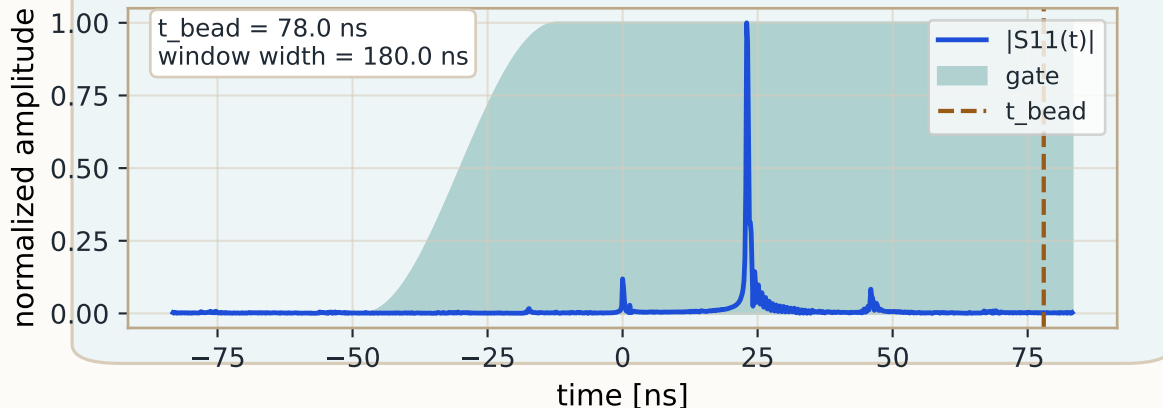
$\Delta\Gamma = S_{11} - [\beta_0 + \beta_1 s]$
Subtract the slow line-by-line background before taking the square root.

Convert To Field

$$E = \sqrt{\Delta\gamma} \frac{C(\nu)}{C(\nu)}$$

$$C(\nu) = \sqrt{\frac{4P_{in}}{\epsilon_0 \alpha_0 i 2\pi \nu}}$$

Example Time Gate At One Aperture Point



Exact Relations Used In The Conversion

$t_{\text{bead}}(x, y, z) = \frac{2}{c} \sqrt{r(z)^2 + x^2 + y^2}$
 $G(t)$ centers a cosine-edged window on the predicted bead return.

$\alpha_0 = \frac{3\Delta\epsilon V_b}{\epsilon_b + 2}$
Before the final conversion, the reduced line products are phase and sign matched to the full-grid reference.